

# Channel Prediction Based on Spatially Dependent Diffusion Models

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**Abstract**—Channel prediction is crucial for addressing the channel aging of time-varying channel in 5G wireless communication. It aims to use past channel state information (CSI) to predict future CSI. Existing channel prediction methods usually process signals based on the temporal correlation of the channels, i.e., considering only the temporal variation relationship of the corresponding elements of the CSI matrix. However, elements of the matrix may be strongly related, which is essential to address channel aging. In this paper, we propose a channel prediction scheme utilizing spatial dependence based on diffusion models (SDDM) by additionally involving the spatial correlation of the channel. Moreover, we regard channel prediction as the denoising process in diffusion models, where instead of directly predicting the actual CSI, we achieve this by predicting the noise injected by the diffusion model. Specifically, the spatial and temporal correlations are extracted by transformers, which serve as the core of the denoising function within the diffusion model. The simulation results demonstrate that the proposed SDDM channel prediction method achieves better performance compared to methods that only involve temporal correlation in most scenarios.

**Index Terms**—Channel prediction, wireless communication, diffusion models, Transformers.

## I. INTRODUCTION

MILLIMETER-WAVE massive multipleinput multiple-output (MIMO) has been a key matter for the fifth-generation (5G) wireless communications. In wireless communication, channel aging refers to the phenomenon where wireless channels change over time. This occurs due to the varying relative positions of users and base stations [1]. This is more pronounced in high-speed mobile environments, because the channel changes faster as the user speed goes higher. Therefore, the accurate prediction is critical to address the issue of channel aging [2]. The challenges of channel prediction arise from the dynamic nature of wireless channels, which are typically affected by multipath fading, Doppler shifts due to mobility, and interference [3]. These make the channel prediction much more complicated, requiring robust method to accurately capture the dynamic variations of channel [4].

Recently, deep learning has been widely used as a solution to address the high complexity and overhead in wireless communication systems, especially for channel prediction tasks [5]–[7]. Denoising Diffusion Probabilistic Models (DDPMs) [8] have been considered among the most powerful generative models, widely applied in computer vision, natural language processing, and large language models. DDPMs learn the data distribution by corrupting clean samples using

additive Gaussian noise and then learning the reverse process to generate new samples from pure noise. As a result, diffusion models exhibit strong noise resistance capabilities, which have been applied to wireless communications, such as channel estimation task [9], [10].

To mitigate the performance degradation caused by channel aging, researchers have extensively studied channel prediction technique, which leverage the temporal correlation of past channels to predict future channels [11]–[13]. Existing channel prediction methods can be broadly categorized into two types: traditional methods and deep learning-based methods.

Traditional methods include the parametric radio channel [14], the basis expansion model [15], and the autoregressive models [16]. These predict channels based on theoretical channel propagation models and estimates of channel parameters such as angle of arrival and departure, and Doppler frequency. However, most real-world propagation channels are nonstationary and do not conform to assumed models, often resulting in suboptimal prediction performance [17].

Deep learning-based methods can be categorized into two approaches. The first is rolling prediction, which uses past channels to predict the channels of the next frame. To predict the second future frame, the predicted channels from the first frame must be combined with past CSI as input for the next iteration. This results in sequential frame-by-frame predictions over time. However, this approach may lead to error accumulation, which impacts performance, especially in mobile millimeter-wave communication. In contrast, the second method is parallel prediction, which uses CSI from past frames to directly predict multiple future frames without iteration. Existing work on parallel prediction is usually based on temporal correlation. However, CSI matrices are in fact not spatially independent, i.e., distinct elements are highly related due to interference [6]. Moreover, channel prediction often performs poorly due to changes in the Doppler effect and user velocity, which poses difficulties for prediction. The denoising capability of diffusion models can effectively address these difficulties. Therefore, unlike the method in [6], which directly predicted future channels using Long Short-Term Memory (LSTM), we view channel prediction as a green denoising process in diffusion models, where instead of directly predicting the actual CSI, we achieve this by predicting the noise injected by the diffusion model.

In order to overcome the performance loss caused by parallel prediction that only considers temporal correlation,

we propose a channel prediction scheme utilizing spatial dependence based on diffusion models (SDDM).

The main contributions of this paper are as follows:

- We use two transformer modules to model both temporal and spatial correlations separately, fully utilizing the multifaceted information of past channels.
- We will use the transformer modules serve as the core of the denoising function within the diffusion model.
- Finally, the simulation results demonstrate that proposed scheme based on diffusion with spatial dependence exhibits better performance in most scenarios.

**Notations:** For clarity, we use bold uppercase letter to denote matrix, bold lowercase letter to denote vector, and non-bold letter to denote scalar.  $\mathbf{x}^{(t_1:t_2)}$  denotes a continuous sequence from time  $t_1$  to  $t_2$ .  $(\cdot)^\top$  is the transpose of a matrix.  $\mathbf{X}[i, j]$  denotes the element at  $i$ -th row and the  $j$ -th column of  $\mathbf{X}$ . The complex number field is represented by  $\mathbb{C}$  and the real number field is represented by  $\mathbb{R}$ .  $\text{vec}(\cdot)$ ,  $\text{dvec}(\cdot)$  denote the vectorization of a matrix and the inverse process, respectively.  $\Re(\cdot)$ ,  $\Im(\cdot)$  denotes the real and imaginary parts of a matrix, respectively. The symbol  $\Theta$  denotes the set of model parameters and  $\mathbb{E}\{\cdot\}$  denotes the statistical expectation.  $[\mathbf{x}, \mathbf{y}]$  denotes concatenate  $\mathbf{x}$  with  $\mathbf{y}$ .

## II. SYSTEM MODEL

In this paper, we consider a model of a MIMO system designed to operate in the Non-Line-of-Sight (NLOS) environment, adhering to the technical specifications of 3GPP TR 38.901. Both the Base Station (BS) and the User Equipment (UE) are equipped with multiple antennas to facilitate spatial multiplexing and enhance the capacity of wireless communication channels. Signal transmission employs frequency-division duplex (FDD) mode, utilizing Orthogonal Frequency Division Multiplexing (OFDM) as the signal waveform to mitigate interference between subcarriers and improve spectral efficiency.

In the FDD mode, which utilizes different frequency channels to simultaneously handle uplink and downlink communication. The data transmission for uplink and downlink is allocated across two distinct frequency bands. A certain frequency separation is maintained between these two bands to prevent interference between the uplink and downlink transmissions. This enables devices to transmit and receive data simultaneously, achieving full-duplex communication, which is particularly useful in cellular networks (such as 4G LTE and 5G) and satellite communications. Denote  $\mathbb{B}_k$  as the set of users in the  $k$ -th region. Then, the channel between the  $u$ -th ( $u \in \mathbb{B}_k$ ) user and BS can be expressed by

$$\begin{aligned} \mathbf{H}_u(f) &= \int_{\theta' \in \mathbb{A}_k} \alpha_u(\theta') \mathbf{a}(\theta') d\theta' \\ &= \int_{\bar{\vartheta}_k^-}^{\bar{\vartheta}_k^+} |\alpha_u(\theta')| e^{-j2\pi \int \tau_u(\theta') + j\phi_u(\theta')} \mathbf{a}(\theta') d\theta', \end{aligned} \quad (1)$$

where  $f$  is the the carrier frequency of the  $u$ -th user [18], while  $|\alpha_u(\theta')|$ ,  $\phi_u(\theta')$  and  $\tau_u(\theta')$  are the attenuation ampli-

tude, the phase shift and the delay of the incident signal ray coming from direction of arrival  $\theta'$ , respectively. Meanwhile,  $\mathbb{A}_k \triangleq [\bar{\vartheta}_k^- - \bar{\vartheta}_k^+]$  is the incident angle spread of the users in the  $k$ -th region with  $\bar{\vartheta}_k^-$  and  $\bar{\vartheta}_k^+$  being the corresponding lower and upper bounds of angle spread [19].

## III. PROBLEM FORMULATION OF CHANNEL PREDICTION

In this section, we explain the time-varying channel prediction problem in wireless communication. We also compare the modeling capabilities of rolling prediction, and parallel prediction.

Channel prediction is usually considered as a time series prediction problem. The channel matrix has the specific array-frequency structure in massive MIMO-OFDM systems. Therefore, the prediction series contains both spatial and temporal correlations. We denote the CSI matrix by  $\mathbf{H}^t \in \mathbb{C}^{I \times J}$ , where  $t$  is the time slot index of the channel,  $I$  and  $J$  respectively represent the number of transmitting antennas and receiving antennas. Thus, the specific formulation of channel prediction could be stated as

$$\hat{\mathbf{H}}^{t+1}, \dots, \hat{\mathbf{H}}^{t+P} = f_{\Theta}(\mathbf{H}^{t-C+1}, \dots, \mathbf{H}^t), \quad (2)$$

where  $\hat{\mathbf{H}}^{t+i}$  denotes the  $(t+i)$ -th predicted CSI,  $C$  denotes the length of the past series,  $P$  denotes the length of the prediction series, and  $f(\cdot)$  denotes an mapping function.

Deep learning channel prediction schemes often cannot receive complex numbers as inputs, so we vectorize the  $\mathbf{H}^t$  matrix as  $\mathbf{h}^t \in \mathbb{R}^K$  ( $K = I \times J \times 2$ ), where  $t$  is the time index of the channel. We then vectorize them and concatenate them into a vector  $\mathbf{h}^t$ , as follows

$$\mathbf{h}^t = [\text{vec}(\Re[\mathbf{H}^t])^\top, \text{vec}(\Im[\mathbf{H}^t])^\top]^\top. \quad (3)$$

Then, we can know that  $\mathbf{h}$ , takes the form of

$$\mathbf{h} = (\mathbf{h}[1], \dots, \mathbf{h}[k], \dots, \mathbf{h}[K])^\top. \quad (4)$$

So the specific expression of channel prediction can be updated from

$$\hat{\mathbf{h}}^{t+1}, \dots, \hat{\mathbf{h}}^{t+P} = f_{\Theta}(\mathbf{h}^{t-C+1}, \dots, \mathbf{h}^t). \quad (5)$$

The core task of channel prediction is to model the mapping function  $f(\cdot)$ . In order to construct  $f$ , researchers have done a lot of work on channel prediction based on deep learning, including rolling prediction and parallel prediction, but most of them only consider temporal correlation and ignore spatial correlation.

**Rolling prediction:** Rolling prediction refers to autoregressive (AR) prediction with output  $\hat{\mathbf{h}}^{t+i} \in \mathbb{R}^K$ . Using a window of past values to predict the next value, we then shift the window and use the predicted value for the subsequent prediction. This process is repeated until  $P$  values are predicted. The window size is set to the length  $C$  of the past channel, as follows

$$\hat{\mathbf{h}}^{t+i} = \sum_{c=1}^C \phi_c \mathbf{h}^{t+c-C+i-1} + b, \quad (6)$$

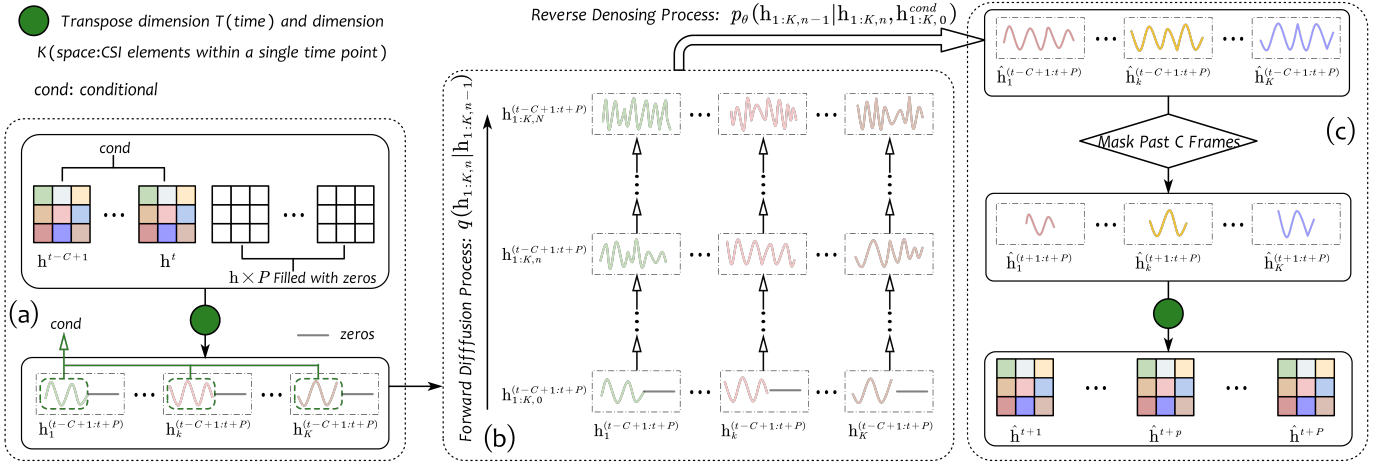


Fig. 1: The overview of the spatially dependent channel prediction scheme based on diffusion models (SDDM), where  $\mathbf{h}^t$  is a column vector, and for ease of representation, we have drawn it as a matrix. The  $\mathbf{h}_{i,n}$  represents adding noise  $n$  steps in the  $i$ -th channel sequence. Figure (b) represents the forward diffusion process in the diffusion model, which adds noise to the input after  $N$  steps to obtain a noise. Figure (b) to (c) is reverse denoising process, which involves training a denoising function  $\epsilon_{\theta}$  and adding conditions (i.e., CSI information from the past  $C$  steps). The denoising function  $\epsilon_{\theta}$  includes serial modeling of the temporal and spatial dimensions of past CSI information to extract spatiotemporal correlations.

where  $i$  from 1 to  $P$ ,  $\phi_i$  represents the coefficient of the  $i$ -th term, and  $b$  represents the bias term. Finally, we obtained all the predicted items  $\hat{\mathbf{h}}^{(t+1:t+P)}$ . The goal of rolling prediction is to find a set of  $(\phi_1, \phi_2, \dots, \phi_C, b)$  that can fit a given set of  $P$  future channels.

**Parallel prediction:** Parallel prediction, refers to using multiple past channels to directly predict multiple future channels, rather than iterating, and its output is  $\hat{\mathbf{h}}^{t+1:t+P} \in \mathbb{R}^{P \times K}$ . The parallel prediction prediction process (5). However, previous parallel predictions often only used temporal correlation, i.e., only modeling the same position of  $\mathbf{h}$ , as

$$\hat{\mathbf{h}}[k]^{t+1}, \dots, \hat{\mathbf{h}}[k]^{t+P} = f_{\Theta}(\mathbf{h}[k]^{t-C+1}, \dots, \mathbf{h}[k]^t), \quad (7)$$

where  $k$  from 1 to  $K$ .

#### IV. SPATIALLY DEPENDENT CHANNEL PREDICTION BASED ON DIFFUSION MODELS

In this section, to alleviate the performance loss caused by spatially independent prediction, we propose a spatially dependent channel prediction scheme based on diffusion models to avoid the propagation of prediction errors and the loss of spatial correlation.

##### A. Spatially Dependent Prediction

Unlike conventional methods that focus only on temporal coherence, spatially dependent prediction method incorporates spatial correlation. This spatial correlation relates to the individual CSI matrix  $\mathbf{H}$ , taking into account the feature correlations among the  $I \times J$  elements within the CSI matrix. Therefore, we decompose the mapping function (5) into  $f_{\Theta_1}$  and  $f_{\Theta_2}$ , which serve as the mapping functions for time and space, respectively. Our first step is also akin to that depicted in (7), which has been updated by

$$\hat{\mathbf{h}}[k]^{t+1}, \dots, \hat{\mathbf{h}}[k]^{t+P} = f_{\Theta_1}(\mathbf{h}[k]^{t-C+1}, \dots, \mathbf{h}[k]^t), \quad (8)$$

where  $k$  from 1 to  $K$ , and  $(\hat{\mathbf{h}}[1:K]^{t+1}, \dots, \hat{\mathbf{h}}[1:K]^{t+P}) \in \mathbb{R}^{P \times K}$ . Then, for the output of the previous step, we will transpose it like

$$(\mathbf{h}[1]^{(t+1:t+P)}, \dots, \mathbf{h}[K]^{(t+1:t+P)}) = (\hat{\mathbf{h}}[1:K]^{t+1}, \dots, \hat{\mathbf{h}}[1:K]^{t+P})^T. \quad (9)$$

Then, we model the spatial correlation by

$$\hat{\mathbf{h}}[1]^{t+p}, \dots, \hat{\mathbf{h}}[K]^{t+p} = f_{\Theta_2}(\mathbf{h}[1]^{t+p}, \dots, \mathbf{h}[K]^{t+p}), \quad (10)$$

where  $p$  from 1 to  $P$ . To achieve the fitting, we ultimately need to transpose the output of  $f_2$  to match the original input format. The goal of channel prediction has become to minimize the loss function

$$\min_{\Theta} \mathcal{L}(\Theta) = \mathbb{E} \left\| \mathbf{h}^{(t+1:t+P)} - \hat{\mathbf{h}}^{(t+1:t+P)} \right\|_2^2, \quad (11)$$

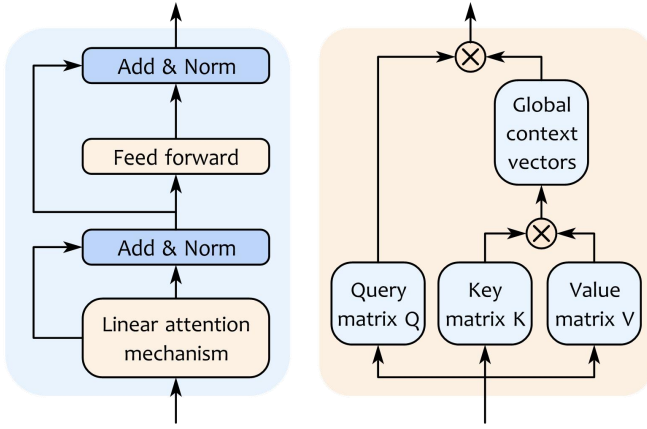
where  $\Theta$  represents the parameters of  $\Theta_1$ ,  $\Theta_2$ , and other components of the model.

##### B. Linear Attention Transformer

Linear attention transformer (LAT) [20] is an enhanced attention mechanism that improves efficiency in processing long sequences by linearizing the computation of attention. The traditional attention mechanism has a time complexity of  $O(n^2)$ . In contrast, LAT reduces the time complexity to  $O(n)$ , which has significant implications for channel prediction.

The LAT achieves this by avoiding the explicit computation of  $QK^T$ , instead leveraging a kernel function  $\phi$  to project the Query and Key into a different space. As shown in Fig. 2, the linear attention mechanism is then computed using the formula

$$\text{Attention}(Q, K, V) = \phi(Q) \cdot (\phi(K)^T \cdot V), \quad (12)$$



(a) Linear Attention Transformer (b) Linear Attention Mechanism

Fig. 2: The structure of Linear Attention Transformer. Subfigure (b) is the specific structure of LAT in subfigure (a). In SDDM, LAT is used as the denoising function  $\epsilon_\theta$  of the conditional diffusion model to extract the temporal and spatial correlations of CSI.

where  $\phi(Q)$  and  $\phi(K)$  are the projections of the query and key, respectively, obtained through the mapping function  $\phi$ . We use linear attention transformer as the  $f_1$  and  $f_2$  functions in (8) and (10), respectively, as

$$f_{\Theta_1} = LAT(P) \text{ and } f_{\Theta_2} = LAT(K), \quad (13)$$

where the  $d$  in  $LAT(d)$  denotes the dimension of input. They are respectively the time layer and the space layer in Fig. 3.

### C. Denoising Diffusion Probabilistic Models

Let us consider learning a model distribution  $p_\theta(\mathbf{x}_0)$  that approximates a data distribution  $q(\mathbf{x}_0)$ . Let  $\mathbf{x}_n$  for  $n = 1, \dots, N$  be a sequence of latent variables from the same sample space of  $\mathbf{x}_0$  (denoted as  $\mathcal{X}$ ). DDPMs [8] are latent variable models of the form  $p_\theta(\mathbf{x}_0) = \int p_\theta(\mathbf{x}_{0:N})$ . It contains two processes, namely the forward process and the reverse process. The forward process is defined by the Markov chain

$$q(\mathbf{x}_{1:N} | \mathbf{x}_0) = \prod_{n=1}^N q(\mathbf{x}_n | \mathbf{x}_{n-1}), \quad (14)$$

where  $q(\mathbf{x}_n | \mathbf{x}_{n-1}) = \mathcal{N}(\sqrt{1 - \beta_n} \mathbf{x}_{n-1}, \beta_n \mathbf{I})$  and  $\beta_n$  is a small positive constant that represents a noise level. DDPMs consider the following specific parametrization of  $p_\theta(\mathbf{x}_{n-1} | \mathbf{x}_n)$  by

$$\mu_\theta(\mathbf{x}_n, n) = \frac{1}{\alpha_n} \left( \mathbf{x}_n - \frac{\beta_n}{\sqrt{1 - \alpha_n}} \epsilon_\theta(\mathbf{x}_n, n) \right) \quad (15)$$

$$\sigma_\theta(\mathbf{x}_n, n) = \sqrt{\tilde{\beta}_n}, \quad \tilde{\beta}_n = \begin{cases} \frac{1 - \alpha_{n-1}}{1 - \alpha_n} \beta_n, & n > 1 \\ \beta_1, & n = 1 \end{cases}, \quad (16)$$

where  $\epsilon_\theta(\cdot, \cdot)$  is a trainable denoising function, as shown in Fig. 3. The reverse process of DDPMs can be trained by minimizing the loss function

$$\mathcal{L}(\theta) = \mathbb{E} \|\epsilon - \epsilon_\theta(\mathbf{x}_n, n)\|^2, \quad (17)$$

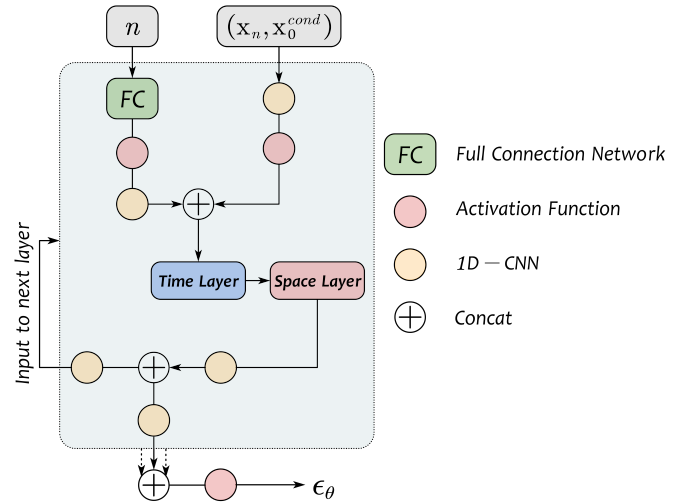


Fig. 3: The structure of the denoising function  $\epsilon_\theta$ . The structures of the *Time Layer* and the *Space Layer* are completely identical, except that their input dimensions are different, with one focusing on the time dimension and the other on the spatial dimension.

where

$$\mathbf{x}_0 \sim q(\mathbf{x}_0), \epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I}), \mathbf{x}_n = \sqrt{\alpha_n} \mathbf{x}_0 + (1 - \alpha_n) \epsilon. \quad (18)$$

### D. Channel Prediction Based on Spatially Dependent Diffusion Models (SDDM) Framework

This section will elaborate on how to construct the LAT and conditional DDPMs to implement the SDDM framework, as well as how to achieve parallel channel prediction based on SDDM.

**Data transposition:** The first dimension of the given CSI data  $\mathbf{h} \in \mathbb{C}^{T \times \frac{K}{2}}$  ( $T = C$ ) is the time axis  $T$ , representing the CSI of each time step of the CSI variation over time. The CSI at each time point is a 2-D matrix, and we pad it with the number ( $P$ ) of CSI matrices to be predicted. Subsequently, it is reshaped into  $\mathbf{h} \in \mathbb{R}^{(C+P) \times K}$  according to (3) and (4), as shown in Figure 1 (a).

**Forward diffusion process:** As shown in Figure 1 (b), the forward diffusion process obtains pure noise by gradually adding Gaussian noise to the original channel. SDDM takes the past  $C$  time steps of CSI as a condition to predict the noise in the diffusion process. Through (17), we will train a denoising function  $\epsilon_\theta$ . As shown in Figure 3,  $\epsilon_\theta$  consists of multiple modules, among which the most important are the time layer and the space layer. The time layer extracts the temporal correlation and the space layer extracts the spatial correlation of the channel through through (13).

**Reverse denoising process:** The denoising process is carried out through a trained denoising function  $\epsilon_\theta$ , which iterates for  $N$  steps in the reverse direction using

$$\mathbf{h}_{n-1} = \frac{1}{\sqrt{\alpha_n}} \left( \mathbf{h}_n - \frac{\beta_n}{\sqrt{1 - \alpha_n}} \epsilon_\theta(\mathbf{h}_n, \mathbf{h}_0^{cond}, n) \right) + \sqrt{\sigma_\theta(\mathbf{h}_n, n)} \mathbf{z}, \quad (19)$$

TABLE I  
DATASET DESCRIPTIONS

| Parameter                  | $v = 30km/h$                       | $v = 60km/h$ |
|----------------------------|------------------------------------|--------------|
| Channel type               | UMa                                |              |
| UE distribution            | outdoor-NLOS                       |              |
| Number of sectors          | 7                                  |              |
| Number of sectors per cell | 3                                  |              |
| Number of UEs per sector   | 10                                 |              |
| Simulation drop            | 5                                  |              |
| Samples Per UE             | 153                                | 308          |
| Total samples              | 160650                             | 323400       |
| Duplex, Waveform           | FDD, OFDM                          |              |
| Multiple access            | OFDMA                              |              |
| Frequency Range            | FR1, 2GHz                          |              |
| Inter-BS distance          | 200m                               |              |
| Channel model              | According to TR 38.901             |              |
| BS Tx power                | 44dBm                              |              |
| BS antenna height          | 25m                                |              |
| UE antenna height & gain   | Follow TR 36.873                   |              |
| Numerology                 | Slot/non-slot: 14 OFDM symbol slot |              |
| SCS                        | 15kHz                              |              |
| Simulation bandwidth       | 52 RB                              |              |
| Subband bandwidth          | 4 RB                               |              |
| Spatial consistency        | Enable                             |              |
| Sample interval            | 5ms                                |              |

where  $n$  from 0 to  $N$ ,  $\mathbf{h}_0^{cond}$  is based on the past CSI as a condition. The process then gradually removes the noise and samples to obtain the final CSI. The resulting CSI includes  $C$  past frames and  $P$  future frames, with the last  $P$  frames being the channel prediction target we need, as shown in Figure 1 (c).

## V. SIMULATION RESULTS

In this section, we will present the simulation scenario and default algorithm parameters. Then, the performance of the SDDM will be evaluated and analyzed.

### A. Simulation Setup

We consider the outdoor massive NLOS scenario that is collected by China Mobile communication platform. Under NLOS conditions, the propagation path between the transmitter and receiver is obstructed by objects such as buildings and trees, preventing direct signal transmission. Instead, the signal reaches the receiver through diffraction, reflection, scattering, and other indirect means. The NLOS scenario typically encompasses a variety of signal measurements across different physical environments, including signal strength, arrival time, and CSI. The dataset consists of two subsets corresponding to speeds of  $30km/h$  and  $60km/h$ , with their specific details presented in Tab. I.

**Baselines:** In this paper, we select four deep learning methods as the baseline models for comparison with SDDM. They are Parallel Transformer [1], Rolling LSTM [5], STNN [6], and Rolling RNN [7], all of which have demonstrated outstanding performance in the channel prediction task. Furthermore, to compare the performance with non-deep learning

methods, we have included four non-deep learning methods. They are also labeled as PVEC [11], AutoRegressive Integrated Moving Average (ARIMA), Kalman Filter, and Fast Fourier Transform (FFT). These methods are training-free and use the rolling prediction approach to sequentially obtain the CSI for each future frame.

**Environment:** All experiments in this paper were implemented using PyTorch, trained using the Adam optimizer, and executed on a single NVIDIA RTX A6000 GPU with 48 GB memory.

### B. Performance of Channel prediction Based on Spatially Dependent Diffusion Models

This subsection validated the effectiveness of the proposed SDDM method, and compare its performance with several other baseline models. This performance is measured by Normalized Mean Square Error (NMSE), where a larger negative value indicates smaller prediction errors and superior model performance. The calculation method for NMSE is

$$NMSE = \frac{\sum_{i=1}^I \sum_{j=1}^J \left\| \mathbf{H}[i, j] - \hat{\mathbf{H}}[i, j] \right\|^2}{\sum_i \sum_j \left\| \mathbf{H}[i, j] \right\|^2}. \quad (20)$$

**Comparison with deep learning methods:** In Fig. 4 (a) and (c), we compare the NMSE performance versus frame between the proposed algorithm with other deep learning algorithms such as the RNN, LSTM and Transformer at  $v = 30km/h$  and  $v = 60km/h$ . Fig. 4 (a) shows the  $P = \{3, 6, 9, 12, 15\}$  are shown according to  $C = 25$ , the user speed  $v = 30km/h$ . Fig. 4 (b) shows the channel prediction results of  $P = \{5, 10, 15, 20, 25\}$  are shown according to  $C = 40$  and the user speed  $v = 60km/h$ . Our algorithm exhibits the state-of-the-art (SOTA) performance most of the time.

**Comparison with non-deep learning methods:** As shown in Fig. 4 (b) and (d), we compared the prediction performance of SDDM with several well-known non-deep learning methods. It can be observed that traditional machine learning methods such as ARIMA, Kalman Filter, and FFT all exhibit prediction performance that is 30% to 60% worse than that of SDDM. This is because these methods are training-free and struggle to capture the complex channel variations under high-speed conditions. Additionally, they focus solely on temporal correlations without considering spatial correlations, which significantly reduces their ability to extract data features. PVEC, a model specifically designed for channel prediction, also underperforms compared to SDDM. Therefore, by considering correlations in both temporal and spatial dimensions, SDDM can effectively adapt to the rapid and complex channel variations in high-speed scenarios.

**Prediction visualization:** The four subgraphs of the Fig. 5 show the results of proposed SDDM, where  $P = 6$ ,  $C = 25$  when the  $v = 30km/h$  and  $P = 10$ ,  $C = 40$  when the  $v = 60km/h$ . Due to the difference in periodicity of the channel data at different speed, we adopt different values for  $C$  and  $P$  under the two speed conditions to enhance prediction

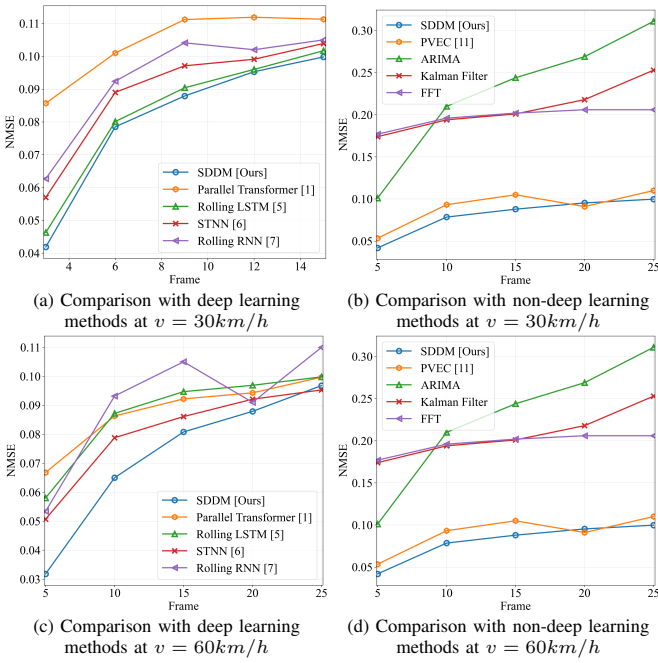


Fig. 4: Comparison of the NMSE results between the proposed channel prediction method SDDM and the baseline models.

accuracy. we can observe that the proposed SDDM channel prediction method could predict future channels with high accuracy.

## VI. CONCLUSION

In this paper, we investigated the time-varying channel prediction problem. We proposed a channel prediction method based on spatially dependent diffusion models. Firstly, by incorporating spatial correlation, we enhanced the robustness of channel prediction, effectively addressing the challenges posed by rapid environmental changes. In addition, by utilizing the diffusion model for channel prediction, we improved the overall prediction quality, demonstrating the effectiveness of our method.

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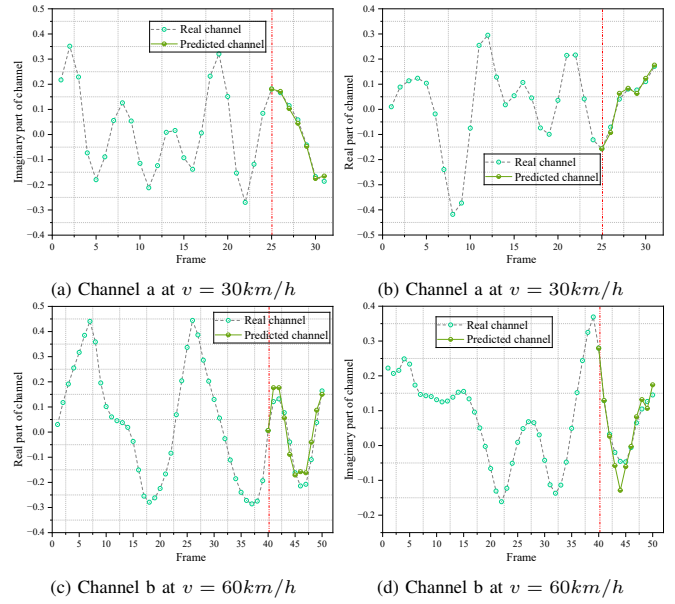


Fig. 5: The channel prediction results of proposed SDDM algorithm. The gray line is the real channel and the green line is the predicted channel. The part in front of the red line is the past part.

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